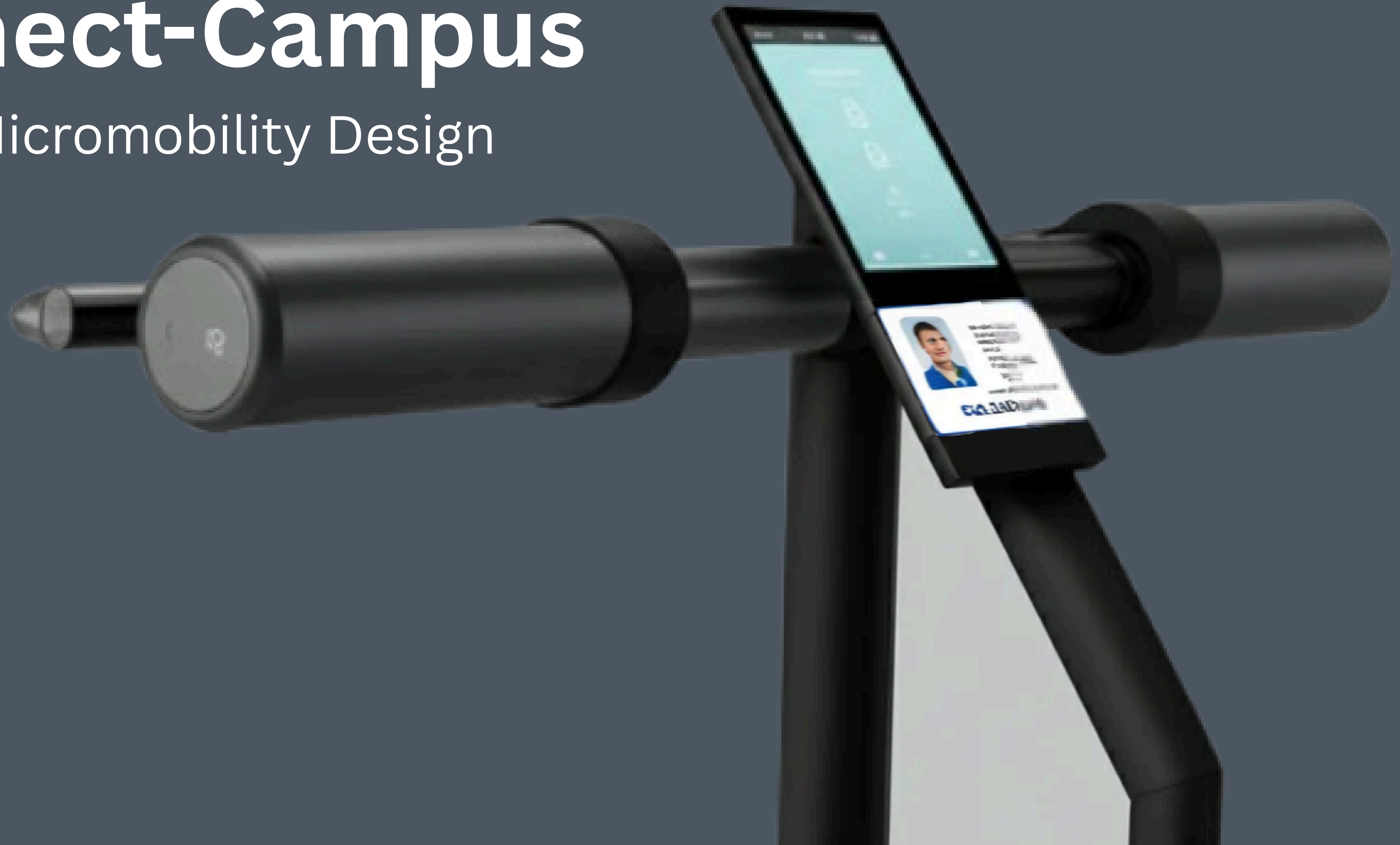


Project: Connect-Campus

A Systems-Based in Micromobility Design



Sri Charan Reddy

The On-Site "Last Mile"

Large campuses are small cities, but they lack smart, on-demand infrastructure. This project proposes an integrated ecosystem to solve the on-site mobility gap.

My Role & Process



Role: Lead Product Designer (Solo Project)



Responsibilities: User Research (Qual/Quant), Market Analysis, Industrial Design (Vehicle/Hub), CMF Strategy, Manufacturing & Cost Analysis.



Process: This case study follows a 4D process: Discover, Define, Develop, and Deliver.

1. Discover & Define

Validating the problem with data and empathy.

Research Methodology

-  **Secondary Research:** Market scan of existing public (Bird, Lime) and private (shuttle) "solutions" to identify feature gaps and business failures.
-  **Primary (Qualitative):** 1-on-1 interviews (n=15) with students & employees to build empathy and uncover deep-seated frustrations.
-  **Primary (Quantitative):** Google Form survey (n=250) to quantify pain points, travel habits, and validate the qualitative findings.

Secondary Research: Market Gaps

Public App-Based

Problem: No fixed parking. Fleets are chaotic, block pathways, and are unreliable. Business model is misaligned with campus needs (e.g., leaving campus).

Private Shuttles

Problem: Inflexible. They run on fixed routes and schedules, failing to provide the on-demand, point-to-point transport that users actually need.

Primary Research: Qualitative Insights



"I have 10 minutes to get from my lab to my lecture. The shuttle takes 15. Walking takes 20. I have to run, and I still show up late and sweaty."

— 3rd Year Engineering Student (Persona: "The Time-Pressed")

Primary Research: Current User Journey Map



Awareness

Class ends in 10 mins.
Next class is 1.5km away.

Anxiety, Stress



Consideration

Check shuttle app. Next
one is in 12 mins.

"Shuttle is useless"



Action

Starts speed-walking or
jogging across campus.

Physical strain



Arrival

Arrives at class 8 minutes
late, out of breath.

Embarrassment



Post-Action

Dreads the next cross-
campus class change.

Low satisfaction

Problem Synthesis & ROI



The Cost of Inefficiency

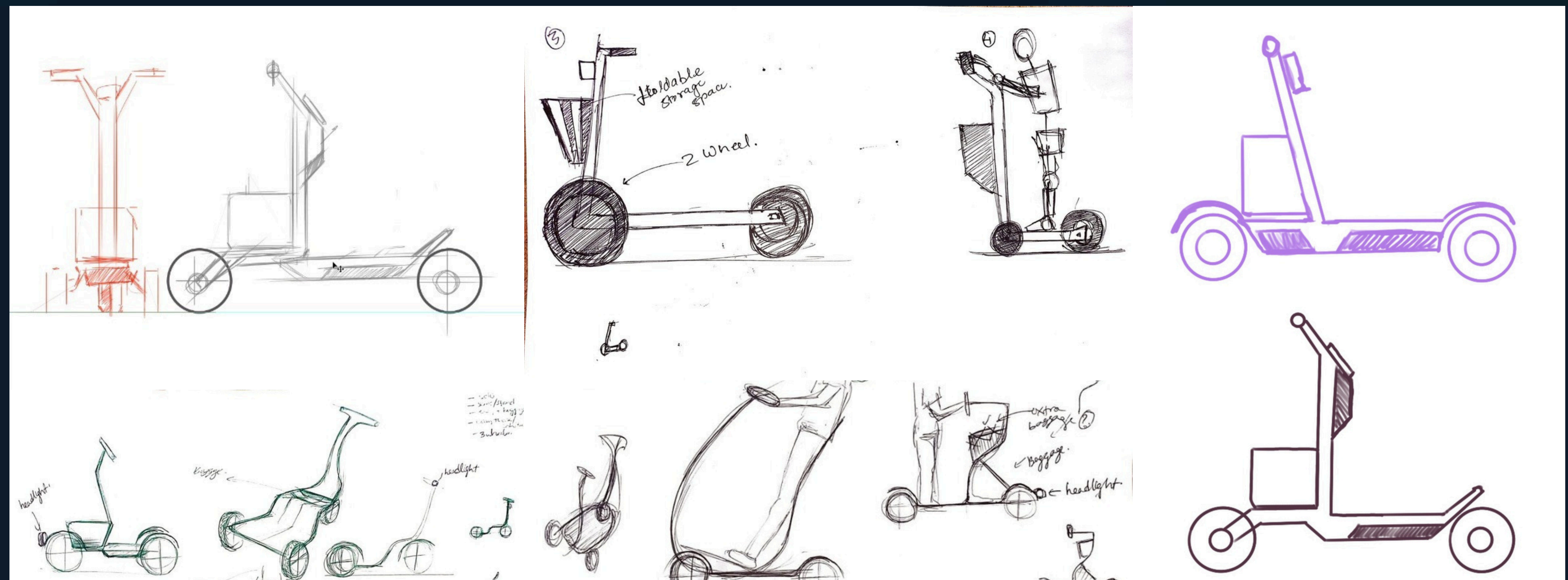
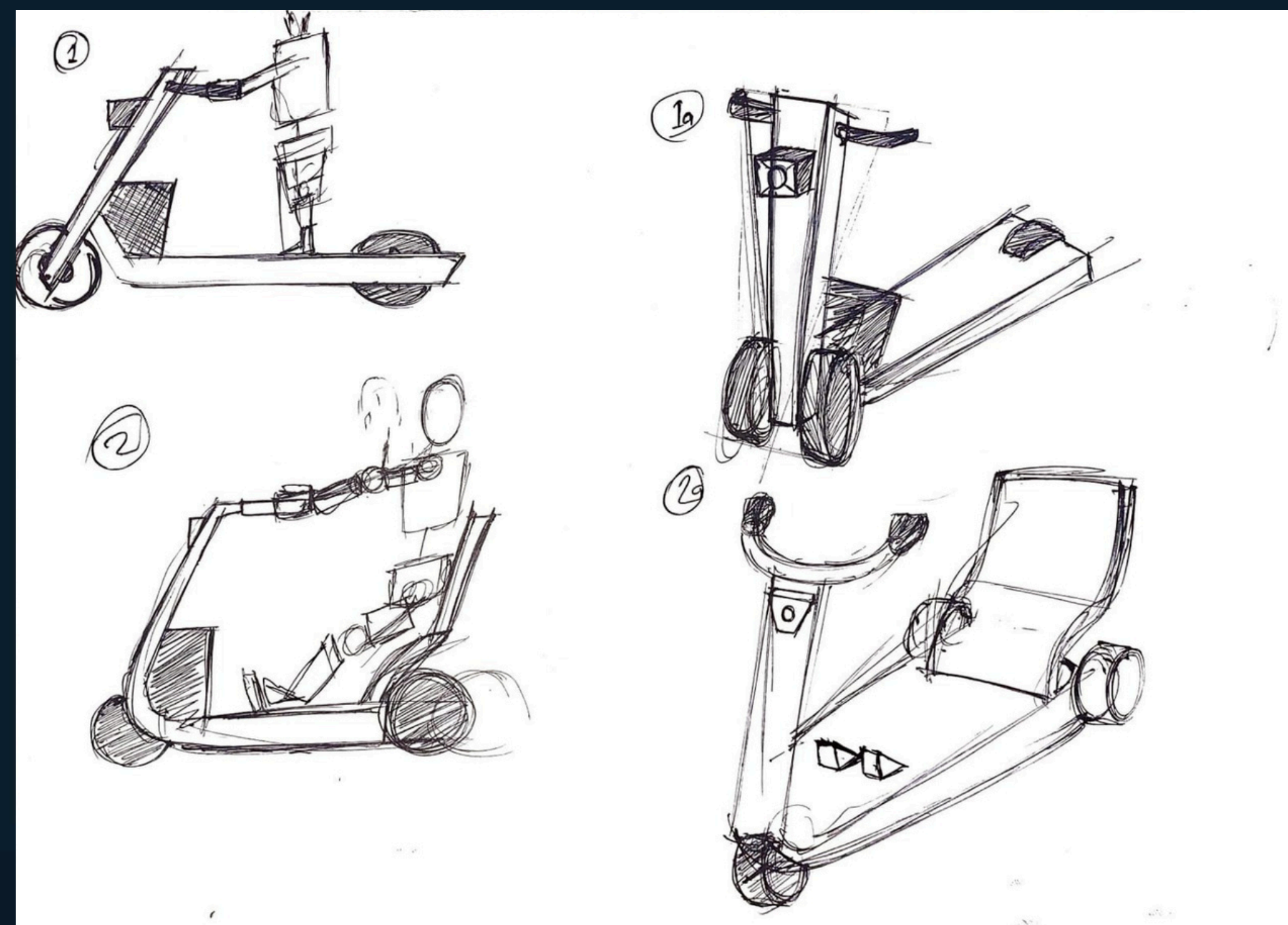
The problem is not just "inconvenience," it's a quantifiable waste of institutional resources.

- ✓ **Time Loss:**
 $(15 \text{ min walk}) \times (2.5 \text{ trips/day}) \times (20,000 \text{ students}) =$
12,500 hours
of wasted time per day.
- ✓ **Asset Underutilization:**
Students and faculty avoid useful buildings (libraries, labs) that are "too far away."
- ✓ **The Core Insight:**
A 5-minute ride replacing a 20-minute walk creates
15 minutes
of productive time, per user, per trip.

2. Develop

From research insights to a tangible, validated solution.

Concept Generation & Iteration



Concept Rejection Rationale

Criteria	Concept A (Pod)	Concept B (Trike)	Concept C (Reverse Triek)
Footprint (Width)	1.5m (Too wide)	1.1m (Too wide)	0.2 (Fits paths)
Pathway Compliance	Fails sidewalk & accessibility path standards.	Fails 0.9m minimum path standard.	Passes all path standards.
Cost (Est. BOM)	\$10,000+	\$3,000+	~\$1,200
Decision	Rejected	Rejected	Selected

The Selected Solution: A Closed-Loop Ecosystem



The Smart Vehicle

A commercial-grade vehicle with onboard IoT and a swappable battery, built for 24/7 uptime.



The Charging Hub

A smart parking and charging station that acts as the 'home' for every vehicle, eliminating chaos.



The Geo-Lock

An intelligent software fence that controls speed, access, and parking for total safety.

System Architecture



How It All Connects

The system is a constant loop of data.

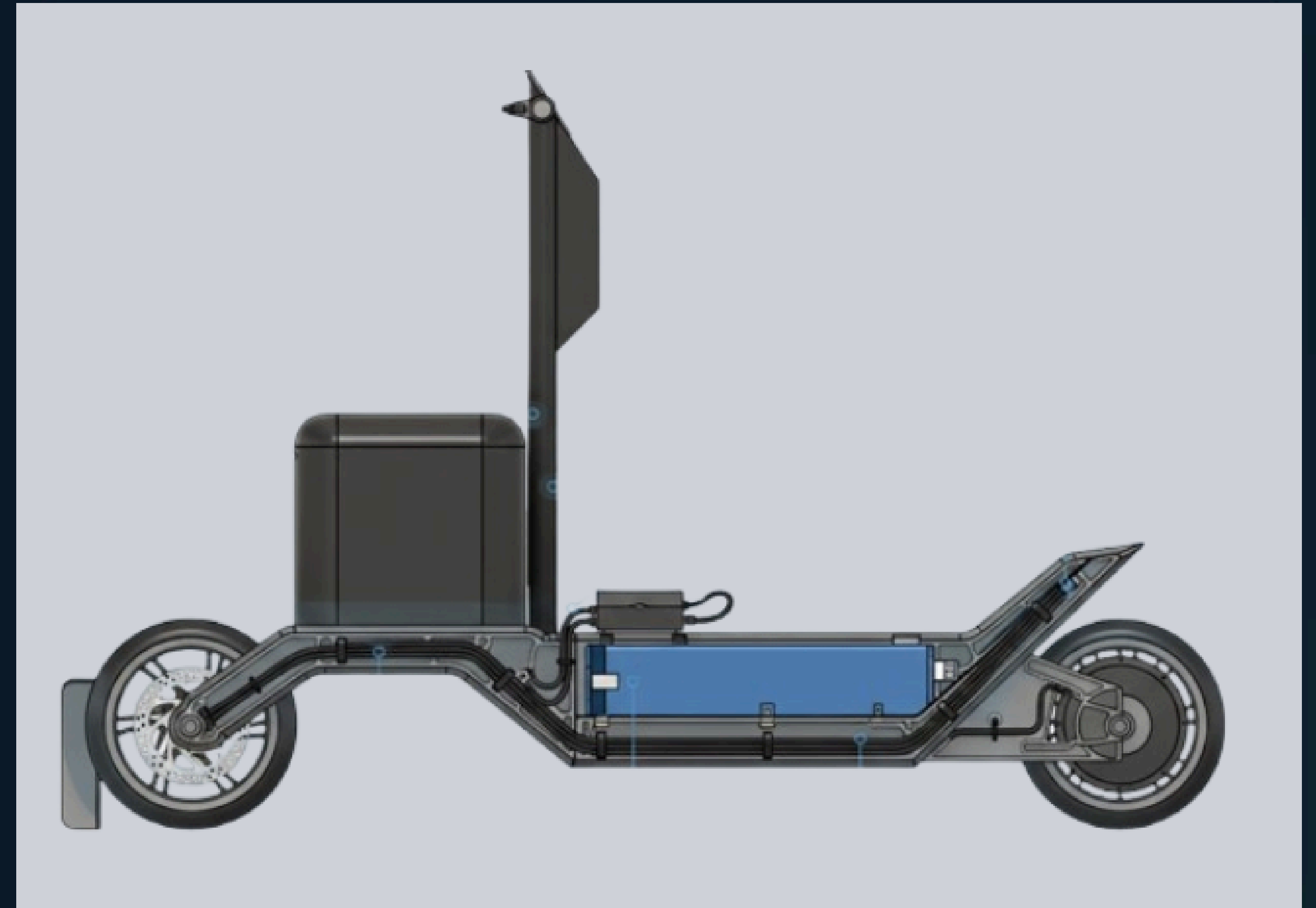
- ✓ **1. User (ID) :**
Requests unlock at a Hub.
- ✓ **2. Cloud Server:**
Authenticates user, sends command to Hub & Vehicle.
- ✓ **3. Hub & Vehicle (IoT):**
Hub unlocks. Vehicle's IoT module activates, tracking GPS & speed.
- ✓ **4. Geo-Lock:**
Vehicle self-enforces "Slow-Go" or "No-Go" zones in real-time.
- ✓ **5. Parking:**
User docks at a new Hub. Hub confirms lock, ends ride, and reports status to Cloud.

Component 1: The Smart Vehicle

Designed for Uptime

This is not a consumer scooter. It is a commercial tool.

- 6061 Aluminum frame
- Swappable 48V battery
- Solid (airless) 10-inch tires
- Onboard 4G IoT & GPS module
- IP67 water resistance



Component 2: The Charging & Parking Hub

The End of Chaos

The hub is the anchor of the system. It's the only place a ride can be ended, ensuring vehicles are always charged and organized.

- Mechanical locking system
- Contact-based auto-charging
- LED status indicators
- Modular (2, 4, or 6-dock config)



3. Deliver (Part 1)

CMF Strategy: Designing for Environment, Durability & Purpose.

CMF: Core Components



Chassis & Frame

Material: 6061 Aluminum Extrusion. **Finish:** Powder Coating. **Why:** Best-in-class strength-to-weight and weather/scratch resistance.



Deck & Grips

Material: High-traction TPE/Silicone. **Finish:** Molded-in texture. **Why:** Maximum grip in all weather; durable and non-porous.



Fenders & Casings

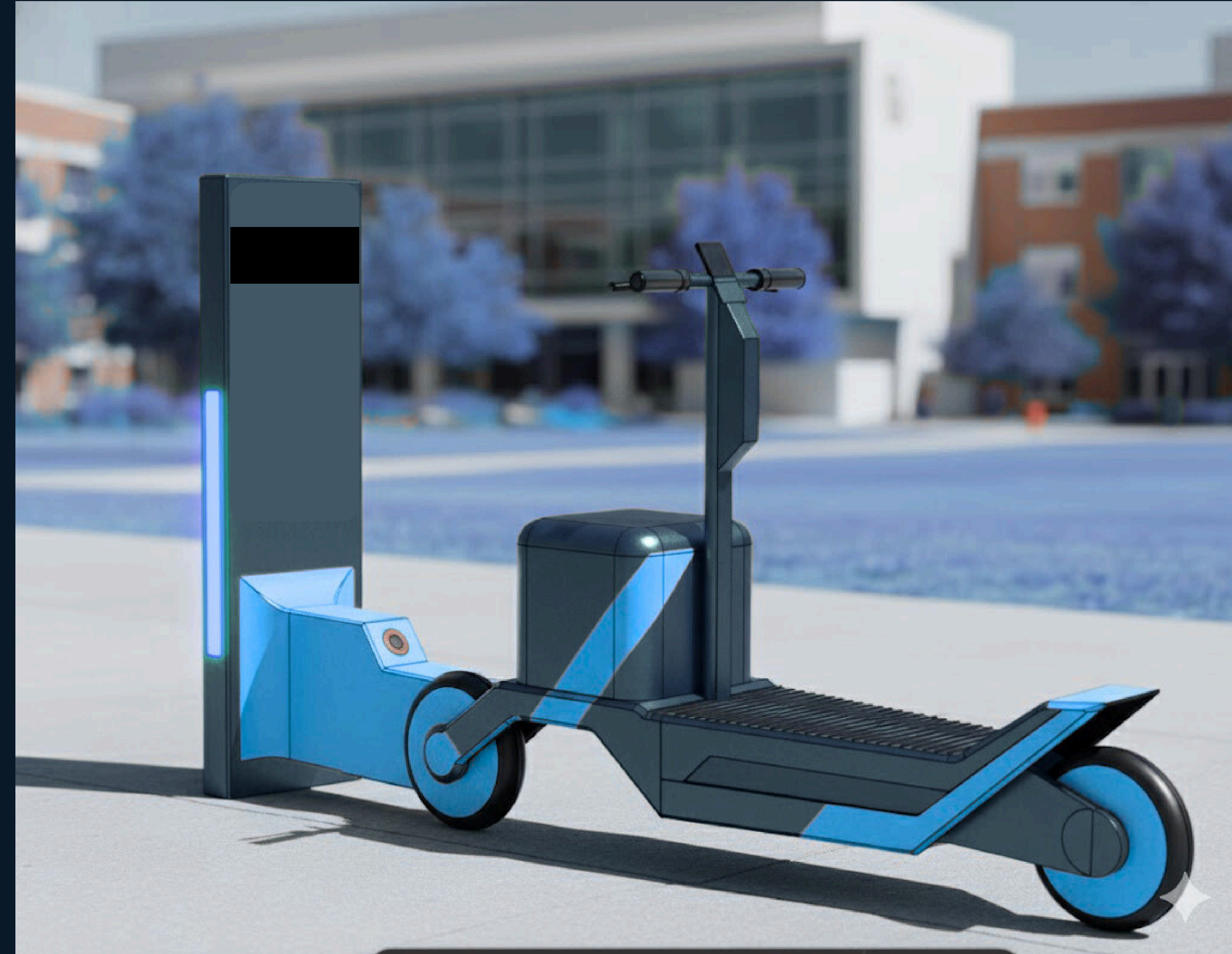
Material: PC/ABS blend. **Finish:** Self-healing matte. **Why:** High impact resistance and a finish that hides minor scuffs.

CMF Persona 1: The University

Focus: Durability & Identity

The university environment is high-use and high-abuse. CMF must be rugged first, but also serve as a point of campus pride.

- **Color:** Bright, bold (school colors). High-contrast accents for visibility.
- **Material:** Thicker-gauge aluminum, impact-resistant PC/ABS.
- **Finish:** Heavy-duty, textured powder coat (hides scratches). Anti-graffiti coating.



CMF Justification: University



Proof of Durability

Why a textured powder coat over a cheaper vinyl wrap?

- ✓ **Abrasion Resistance:**
Powder coat (80-120mg loss) vastly outperforms vinyl (200mg+ loss) in Taber Abrasion Tests. It won't peel or scuff as easily.
- ✓ **Impact Resistance:**
Powder coating is flexible and bonds to the metal, resisting chipping from drops and impacts.
- ✓ **Cost/Benefit:**
Higher upfront cost saves thousands in yearly refurbishment and fleet downtime.

CMF Persona 2: The Corporate Campus

Focus: Professionalism & Integration

The vehicle must blend into a professional environment. It should look like part of the corporate architecture—sleek, modern, and high-tech.

- **Color:** Muted, sophisticated (anodized silver, graphite, deep blue).
- **Material:** Anodized aluminum, premium TPE grips.
- **Finish:** Brushed/anodized finish on metal, soft-touch matte on handles.

CMF Persona 2: The Corporate Campus Focus: Professionalism & Integration



CMF Justification: Corporate

Criteria	Option 1: Brushed Steel	Option 2: Anodized Aluminum	Decision
Cost	Low (\$)	High (\$\$\$)	Anodized. The premium, non-oxidizing finish is a key brand differentiator for the corporate environment.
Weight	Heavy	Light (6061)	
Finish Integrity	Surface finish. Can rust/oxidize.	Integrated (electrochemical). Will not oxidize or peel.	
Perceived Value	"Standard"	"Premium" (Apple-like)	

CMF Persona 3: The Hospital Premises

Focus: Sterility & Calm

In a health environment, CMF must prioritize hygiene. Surfaces must be non-porous, easy to clean, and not contribute to visual stress.

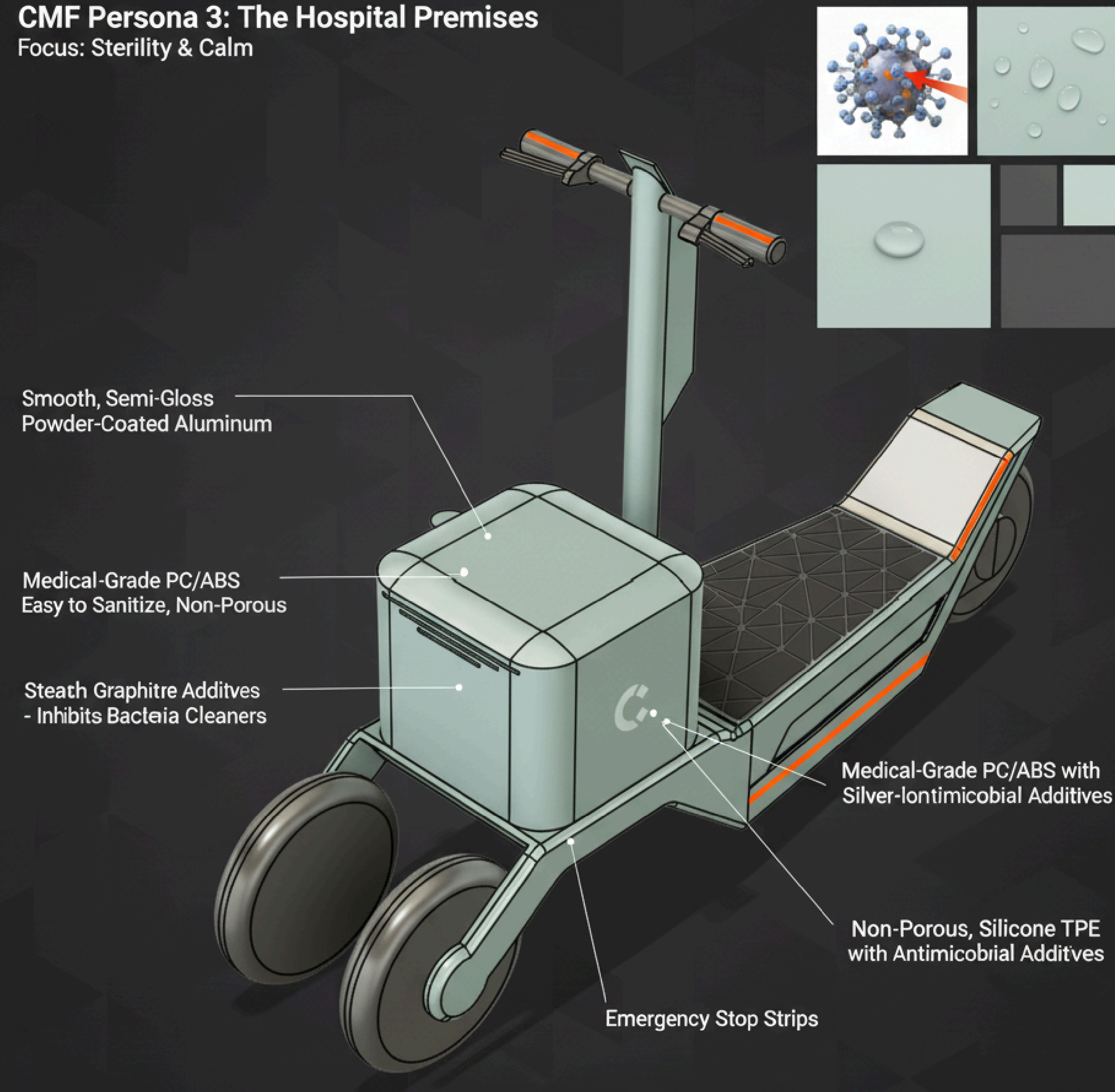
- **Color:** Calming, clean (light blues, greens, sterile white).
- **Material:** TPE/Silicone with antimicrobial additives.
- **Finish:** Smooth, semi-gloss. No deep textures where microbes can hide.



CMF Justification: Hospital

CMF Persona 3: The Hospital Premises

Focus: Sterility & Calm



Designing for Sterility

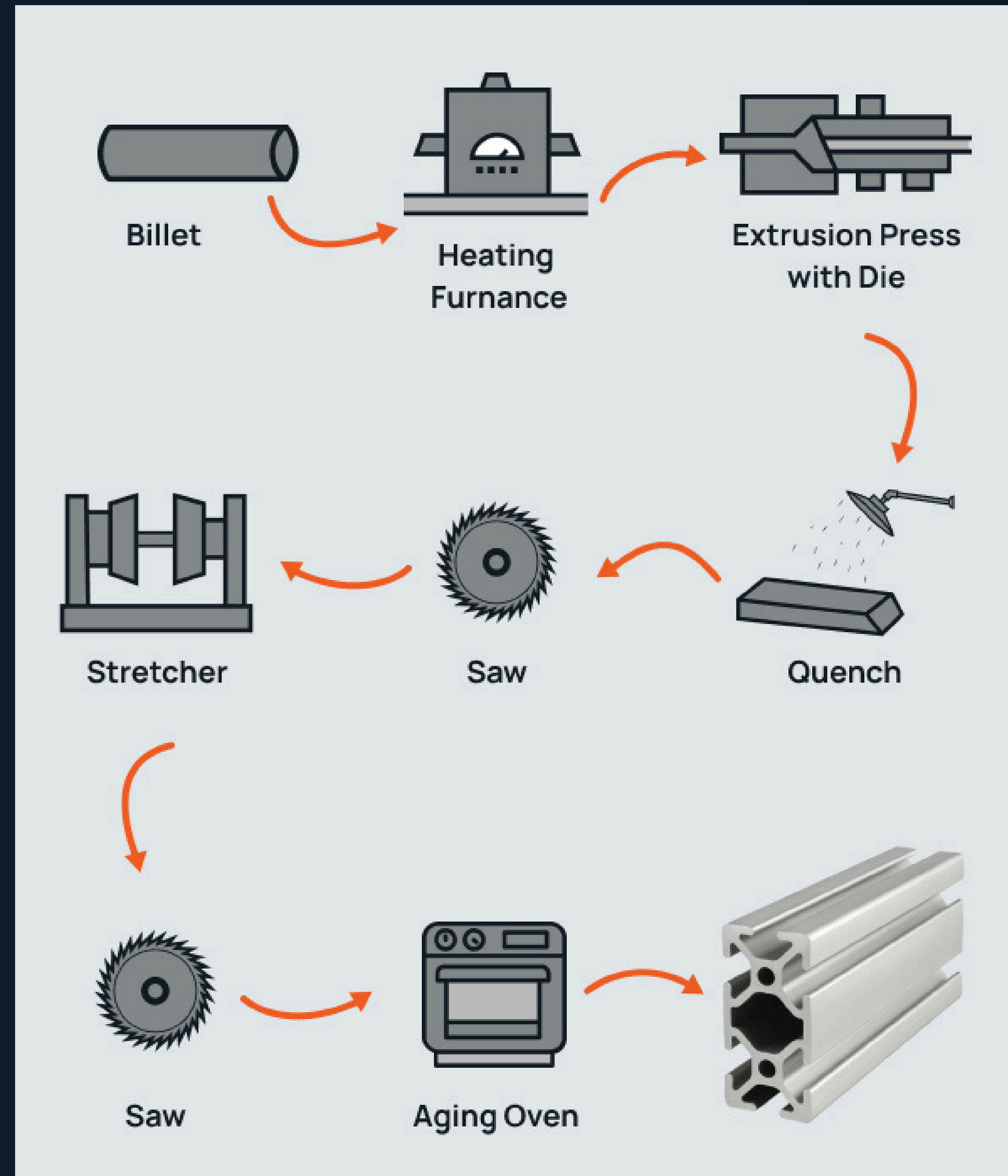
This is a non-negotiable CMF requirement for healthcare.

- ✓ **Material Spec:**
Medical-grade TPE resin with an embedded **silver-ion (Ag⁺) antimicrobial additive**.
- ✓ **Standard:**
Designed to meet **ISO 22196**
(Measurement of antimicrobial activity on plastics).
- ✓ **Mechanism:**
The silver ions are released by ambient moisture and disrupt bacteria cell walls, inhibiting growth 24/7.
- ✓ **Finish:**
All surfaces are smooth (max 2% gloss) and non-porous to comply with hospital "wipe-down" cleaning protocols.

4. Deliver (Part 2)

Manufacturing & Cost: Linking CMF to Production.

Manufacturing: Frame Process



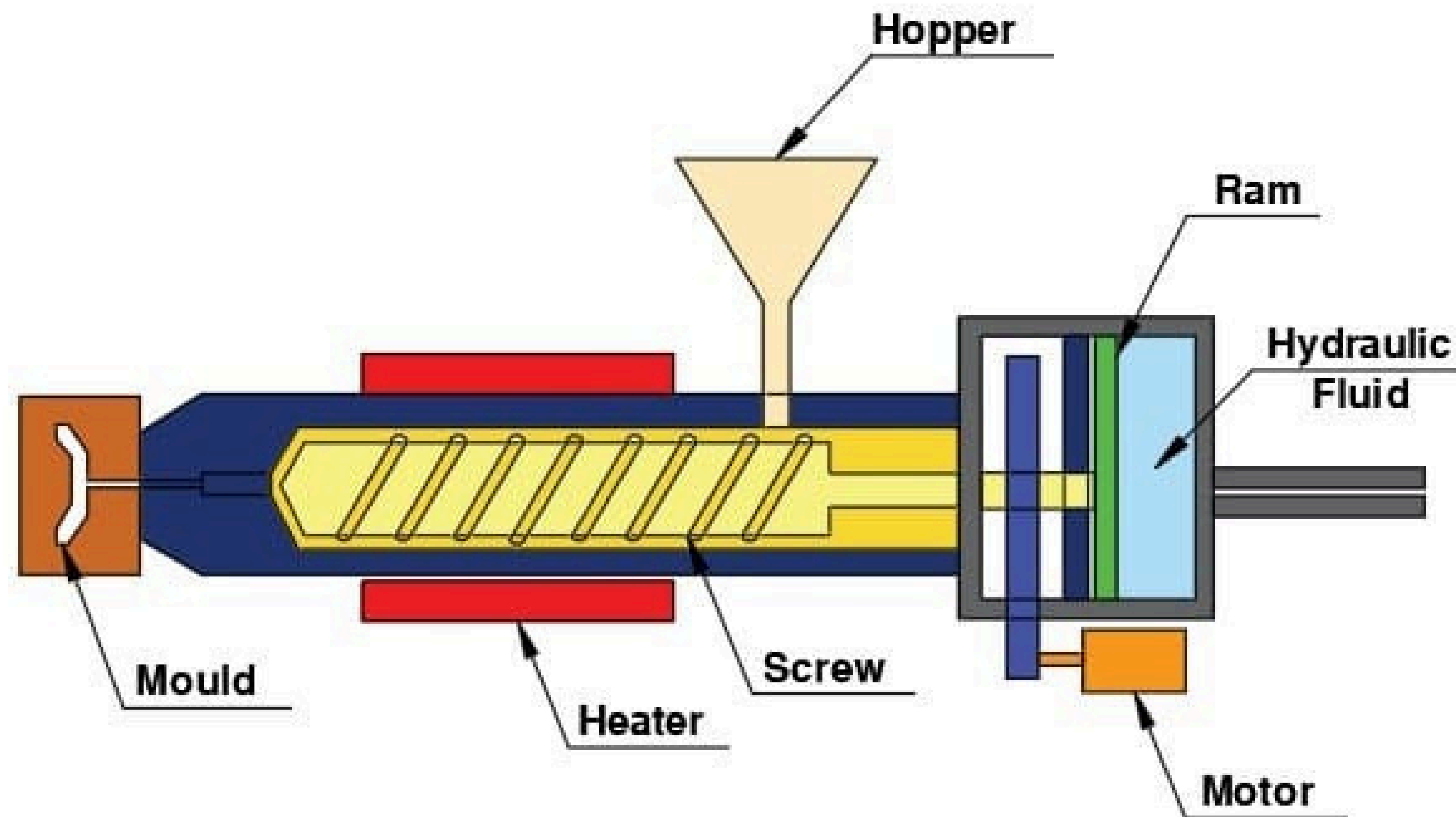
Process: Aluminum Extrusion

Why extrusion over a welded frame?

- ✓ **Strength:**
A single, continuous piece of 6061-T6 aluminum is far stronger than a multi-piece welded assembly.
- ✓ **Tolerance:**
Eliminates weld warping and tolerance stack-up, ensuring perfect alignment of components.
- ✓ **Cost:**
High initial tooling cost, but a very low per-unit cost at scale, making it ideal for a fleet.
- ✓ **Features:**
Allows for integrated features like screw-channels and wire-runs directly in the frame profile.

Manufacturing: Casings & CMF

Injection Molding



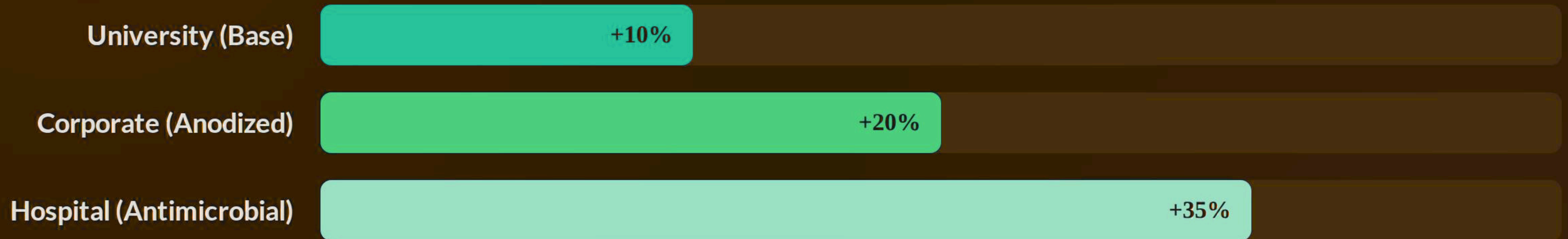
IQSdirectory.com

Process: Injection Molding

CMF is not just applied **to** the part; it **is** the part.

- ✓ **Material:**
PC/ABS pellets are mixed with colorant, UV stabilizers, and (for Hospital) the antimicrobial additive resin **before** molding.
- ✓ **Finish:**
The matte or semi-gloss finish is etched directly into the steel mold (A-B-C finish), ensuring perfect consistency.
- ✓ **Efficiency:**
Color, finish, and material properties are all achieved in a single ~60-second cycle time.

CMF Impact on Per-Unit Cost



Why +35% for Hospital? 10% = Cost of antimicrobial additive resin. 15% = Cost of certified medical-grade polymers. 10% = Slower mold cycles & stricter QC.

5. Validation & Iteration

Failure is the fastest path to a robust product.

Final Design Renders



Vehicle docked in the modular hub.

Rechargeable Battery

Handlebar UI and scan point.

Thank You

Questions?